

CONNECTION DESIGN ADVISOR

Project: Quick Analysis

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Model: STRUCTMIND CORE · PRO

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Greetings. I am STRUCTMIND AI. Based on the provided 3-storey SMF (Special Moment Frame) + OCBF (Ordinary Concentrated Braced Frame) mixed system, I will provide optimal connection recommendations. My analysis prioritizes seismic performance, code compliance (AISC 360-22 / 341-22), and erectability.

The mixed-system approach implies SMF lines will act as the primary ductile system for drift control and energy dissipation, while the OCBF lines will provide significant stiffness and strength. Connections must be designed accordingly.

1. Beam-Column Interior Joints (SMF Line)

These connections are critical for the ductile performance of the moment frame and must accommodate large inelastic rotations.

• 1.1. Recommended Connection Type:

Bolted Flange Plate (BFP) Moment Connection. This is a prequalified connection per AISC 358-22. It is chosen for its high ductility and superior constructability, as it eliminates field welding of the moment-resisting elements.

• 1.2. Governing Limit States (AISC 360-22, Chapter J):

- Shear Transfer (via Shear Tab):
 - Bolt Shear Strength (J3.6)
 - Bolt Bearing/Tearout on Beam Web and Plate (J3.10)
- Shear Yielding of the Plate (J4.2(a))
- Shear Rupture of the Plate (J4.2(b))
- Block Shear Rupture of the Plate (J4.3)
- Weld Strength (Shear Tab to Column Flange) (J2.4)
- Moment Transfer (via Flange Plates):
 - Tensile Yielding of the Flange Plates (J4.1(a))
 - Tensile Rupture of the Flange Plates (J4.1(b))

- Block Shear Rupture of the Flange Plates (J4.3)
- Bolt Shear Strength (J3.6)
- Bolt Bearing on Flange Plates and Beam Flange (J3.10)
- Seismic Checks (AISC 341-22):
- Connection must be designed for the Probable Maximum Moment (M_{pr}) of the beam.
- Column Panel Zone Shear Strength (E3.6f) is often a governing check.
- 1.3. Bolt / Weld Sizing Guidance:
- Bolts: Use high-strength ASTM F3125 Gr. A490 bolts for the flange plates. These must be installed as pretensioned and slip-critical to prevent slippage under load reversal. Shear tab bolts can be Gr. A325 installed snug-tight.
- Welds: The shear tab-to-column flange connection should be made with a CJP (Complete Joint Penetration) weld or double-sided fillet welds sized for the full shear demand. These are considered "demand critical" welds requiring stringent inspection.
- 1.4. Constructability Score & Remarks:
- Score: 8/10
- Remarks: Excellent choice for speeding up erection. All field work is bolting. Fabrication is straightforward but requires good tolerance control for bolt hole alignment. Eliminates costly and time-consuming field CJP welding and associated inspections.
- 1.5. Alternate Solutions:
- Reduced Beam Section (RBS): The classic "dog-bone" cut. Excellent, well-documented performance. Requires more precise fabrication and can sometimes interfere with floor decking or MEP.
- Welded Unreinforced Flange - Welded Web (WUF-W): A fully welded option. Very robust but has the lowest constructability score due to extensive field welding, inspection (UT), and weather sensitivity.

2. Brace-to-Gusset Connection (OCBF Line)

This connection's primary function is to transfer large axial forces (tension and compression) from the brace into the frame.

- 2.1. Recommended Connection Type:

Bolted Brace to Welded Gusset Plate. An HSS brace is bolted directly to a gusset plate, which is in turn welded to the beam and/or column. This is the industry standard for efficiency and reliability.

- 2.2. Governing Limit States (AISC 360-22, Chapter J):

- Brace: Tensile Yielding (D2a), Tensile Rupture (D2b), Block Shear Rupture (J4.3), Compressive Strength (E3/E4).
- Gusset Plate:
 - Whitmore Section Yielding and Buckling (J4.4)
 - Block Shear Rupture at the brace interface (J4.3)
 - Gusset Plate Edge Buckling (AISC Design Guide 29)
- Bolts: Bolt Shear Strength (J3.6), Bolt Bearing/Tearout on Brace and Gusset (J3.10).
- Welds (Gusset to Frame): Weld strength (J2.4), designed to resist the force resultant determined by the Uniform Force Method (UFM).
- 2.3. Bolt / Weld Sizing Guidance:
 - Bolts: ASTM F3125 Gr. A325 bolts are typically sufficient. Since this is a bearing connection, they can be installed snug-tight. Bolt layout should be compact to minimize gusset size.
 - Welds: Use fillet welds to connect the gusset to the beam/column flanges. Weld size is often 5/16" or 3/8" and can be determined directly from the UFM analysis.
- 2.4. Constructability Score & Remarks:
 - Score: 9/10
 - Remarks: Highly constructible. This is a standard, repeatable connection that fabricators and erectors are very familiar with. The main consideration is providing adequate clearance for tightening the bolts.
- 2.5. Alternate Solutions:
 - End Plate Brace Connection: Can simplify gusset design but requires more fabrication on the brace itself and tighter tolerances. Not as common for standard OCBF systems.
 - Pinned Connection: Using a single large pin can be effective but is generally more complex and expensive to fabricate.

3. Column Base Plates

The base plate design depends entirely on its location within the mixed system.

- 3.1. Recommended Connection Type:
 - SMF Columns: Fixed Moment Base Plate. Requires a thick base plate, a large pattern of heavy anchor rods, and often shear keys to transfer moment, shear, and axial load into the foundation.
 - OCBF/Gravity Columns: Pinned Base Plate. A simpler connection designed primarily for shear and axial load, with minimal moment resistance. Anchor rods are typically clustered near the column web.
- 3.2. Governing Limit States (AISC 360-22, J8 & ACI 318-19, Ch. 17):

- Fixed Base:
 - Concrete Bearing Strength (J8)
 - Base Plate Bending at the faces of the column/stiffeners (AISC DG 1)
 - Anchor Rod Tensile Strength (ACI 318)
 - Concrete Breakout/Pullout Strength of Anchors (ACI 318)
 - Weld Strength (Column to Base Plate) (J2.4)
- Pinned Base:
 - Concrete Bearing Strength (J8)
 - Anchor Rod Shear Strength (ACI 318)
 - Concrete Edge Breakout in Shear (ACI 318)
- 3.3. Bolt / Weld Sizing Guidance:
 - Anchors: Use ASTM F1554 Gr. 55 or 105 for fixed bases to provide high tensile strength in a smaller diameter. Use double nuts (one for leveling, one for tightening) for moment transfer. Gr. 36 is often sufficient for pinned bases.
 - Welds: CJP or double-sided PJP welds are required for column-to-base plate connections at SMF columns. Fillet welds are typically sufficient for pinned bases.
- 3.4. Constructability Score & Remarks:
 - Fixed Base Score: 6/10
 - Pinned Base Score: 9/10
 - Remarks: Fixed bases are challenging. Anchor rod placement is critical and unforgiving. Leveling and grouting thick plates is a slow, precise operation. Pinned bases are simple and fast.
- 3.5. Alternate Solutions:
 - Proprietary Anchoring Systems: Some systems (e.g., Peikko, Hilti) offer pre-engineered solutions that can improve tolerance control and speed up installation, but may come at a premium cost.

4. Column Splices

Splices are required to join column sections between floors. Their design is driven by constructability and the forces they must transfer.

- 4.1. Recommended Connection Type:

Bolted Flange and Web Plate Splice. Plates are shop-welded to the bottom of the upper column segment and field-bolted to the top of the lower segment. The splice is typically located ~4 ft above the finished

floor for ease of access.

- 4.2. Governing Limit States (AISC 360-22, Chapter J):

- Bolt Shear Strength (J3.6)
- Bolt Bearing on Plates and Column Flanges/Web (J3.10)
- Gross and Net Section Yielding/Rupture of Splice Plates (J4.1, J4.2)
- Block Shear Rupture of Plates and Column Sections (J4.3)
- Compressive Bearing Strength of Column Ends (J1.4(c)) (if not a complete tension splice).

- 4.3. Bolt / Weld Sizing Guidance:

- Bolts: ASTM F3125 Gr. A325 bolts installed in a bearing-type connection are usually sufficient. In SMF columns, design forces should consider potential moments from frame action.

- Location: Placing the splice 4 ft to 6 ft above the floor keeps workers off ladders and allows for simple, safe connection work.

- 4.4. Constructability Score & Remarks:

- Score: 8/10

- Remarks: Very common and efficient. The use of erection aids like guide lugs is highly recommended. Requires coordination to ensure crane can place the upper column segment while erectors install bolts.

- 4.5. Alternate Solutions:

- Directly Welded Splice: Provides a clean, flush aesthetic but is very expensive and slow due to field welding, NDT inspection, and weather delays. Generally avoided in commercial construction.

- Proprietary Splices: Some systems offer quick-connect mechanical splices, but they are less common and may have specific design limitations.